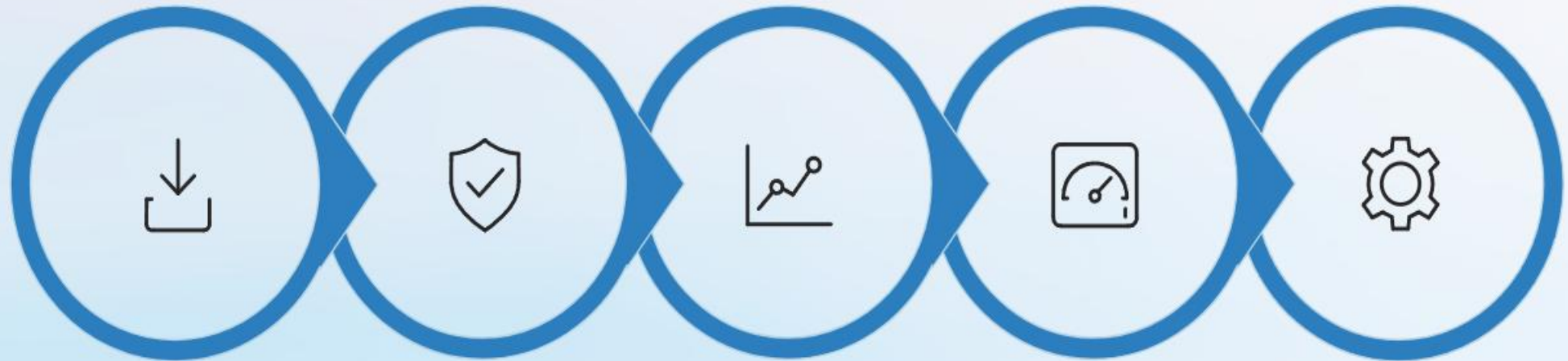


Course Evaluation Data Pipeline

From Raw data to Agentic Ai insights



Raw Ingest

**Bronze
Validate**

**Silver
Analytics**

Gold Outputs

**Prod
Controls**

University course evaluation analytics, privacy protection, and AI-powered insights.

Initial Thought Process

Goal for this project:

- This is **not** mainly a **dashboard problem**.
- The real problem is turning a large flat feedback file into **trustworthy academic insight**.
- AI should not read the raw data directly.
- First: validate, clean, protect privacy, structure the data, and define reliable metrics.
- Then: AI answers decision questions through controlled tools and trusted views.
- Every AI answer must include evidence, sample size, confidence, and limitations.
- How I Can make AI to reduce hallucination, sending data to ai, not more not less as required.
- Is the **data trustworthy? system scale? right visualizations?**

Data Analysis

- Dataset is a structured CSV/Excel feedback file.
- Sample has 224 anonymized rows.
 - Input rows: 224, Input columns: 32
- Rating fields:
 - i1-i7 = instructor feedback
 - c1-c5 = course/curriculum feedback

Valid rating counts:

- i1, i2, i3, i7 = 224 valid values
- i4, i5, i6 = 223 valid values
- c1, c2, c3 = 224 valid values
- c4, c5 = 223 valid values
- Invalid non-numeric ratings: 0
- Out-of-range ratings: 0
- Rows with at least one missing rating: 3

Key Missing Values:

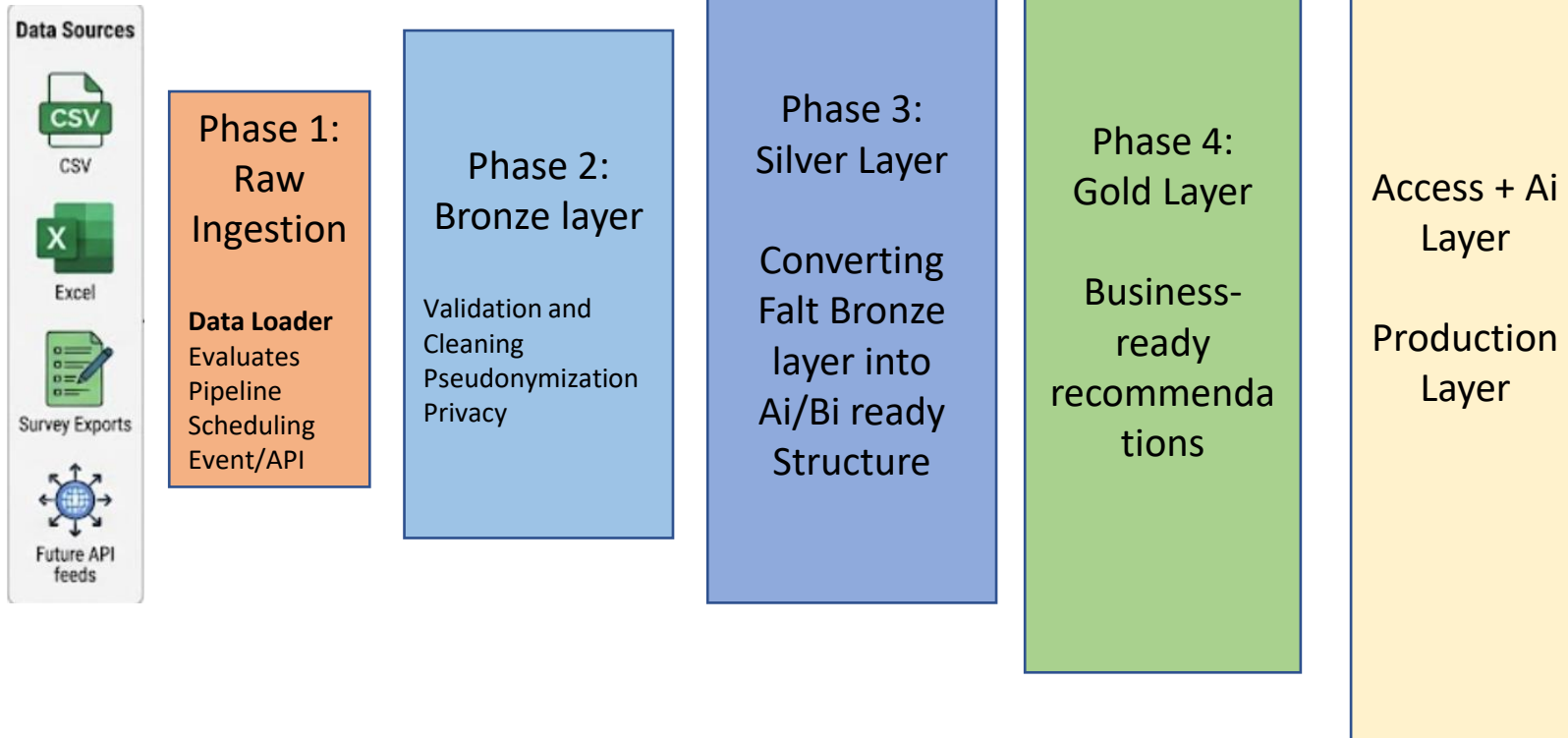
LocationCode: 224 missing

NumberCode: 222 missing

Comments are stripped in the sample.

Environment	Pipeline Orchestration	Storage	Analytics
MVP / Local	Mage AI	PostgreSQL / local files	SQL views
AWS	AWS Glue / Step Functions	S3	Athena / Redshift
Google Cloud	Cloud Composer / Dataflow	Cloud Storage	BigQuery
Azure	Azure Data Factory	ADLS Gen2	Synapse / Fabric

Methodology



Challenges

1. Statistical Integrity & Normalization

- **Distribution (Spread)** - Raw averages can be misleading.
- **Low sample sizes** need confidence labels.
- **Easy Grader bias**- Skipped questions must not be treated as zero.
- **Momentum** - Instructor quality and course design can be confused.

2. Security & Compliance

- AI can hallucinate if it reads raw data directly. Flat CSV is not enough for reliable AI reasoning.
 - **Data Lineage** - Comments can contain private information.
 - **PII Masking**, Access Control, AI Safeguards, Schema

1. Statistical Integrity & Normalization Solutions

- Distribution (Spread) → Calculate Variance & Standard Deviation
- Low Sample Sizes → Apply a Minimum Threshold & Bayesian Smoothing
- Easy Grader Bias → Z-Score Normalization
- Skipped Questions → Exclude from Denominator (Use `NULL` , not `0`)
- Momentum → Time-Series Cohort Analysis

2. Security, Compliance & AI Solutions

- AI Safeguards → Guardrails & RAG Architecture
- Access Control → Row-Level Security (RLS)
- PII Masking → Named Entity Recognition (NER) Filtering
- Data Lineage → Implement Metadata Tagging
- Schema Limitations → Star Schema Semantic Layer

Demo



DATA ENGINEERING
[RAW DATA – CLEANED DATA]



AI ASSISTANT OVER CURRENT
DASHBOARD CONTEXT



ENVIRONMENT STETUP

Findings and Results

Transforming Raw Student Feedback into Privacy-Safe, AI-Ready Decision Intelligence

1. Data Architecture & Pipeline

- **Medallion Pipeline:** Designed a robust multi-stage framework ([Raw](#) → [Bronze](#) → [Silver](#) → [Gold](#))
- **The Star Schema Shift:** Converted flat Bronze data into a Silver Star Schema to enable clean, multi-dimensional query processing.
- **Engineered Insights:** Built dedicated Gold analytical views for [Course Health](#), [Professor-Course Fit](#), and [Curriculum Reviews](#).

2. Data Governance & Privacy Engineering

- **Automated Data Quality:** Enforced structural schema validation, strict rating boundaries, custom NULL handling, and automated metric checks.
- **Identity Anonymization:** Obfuscated sensitive administrator views by replacing raw names with stable instructor pseudonyms and aliases.
- **PII Redaction Pipeline:** Intercepted free-form text blocks using Microsoft Presidio and NER models to scrub personal identifiers from [itxt](#) and [ctxt](#).

3. AI Safety & Semantic Guardrails

- **Deterministic SQL Guardrails:** Restricted LLM text-to-SQL generation to validated views to prevent analytical hallucinations.
- **Strategic RAG Separation:** Confined Retrieval-Augmented Generation ([RAG](#)) strictly to redacted text comments; barred AI from raw numeric calculations.
- **Structured AI Context:** Used relational table bounds so the AI explicitly isolates curriculum issues from instructor performance.

4. API Delivery & Interface Results

- **Unified Access Layer:** Engineered a high-performance FastAPI [backend](#) to securely stream processed metrics to downstream BI tools.
- **Operational Prototype:** Built an interactive [dashboard](#) serving as a live validation prototype and analytical interface.
- **Large Scale Analytics:** Turned structured text-and-csv file into an enterprise-grade, [privacy](#)-compliant analytical engine.

CONCLUSION

- AI is useful only after the data is trustworthy.
- Structured Approach: validate data → protect identities → model and Schema → expose trusted views.
- AI should answer through controlled tools, SQL guardrails, approved views, and RAG only for redacted comments.
- Final recommendations must include evidence, response count, confidence, and limitations.





THANK YOU

ANY QUESTIONS?
